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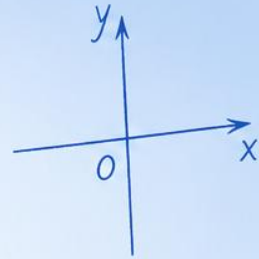
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$$ax + by = c$$
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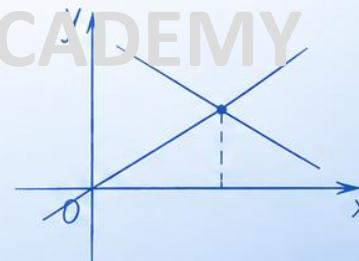


CHAPTER 1

Pairs of Equations

$$\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases}$$

MASTER X'S ACADEMY



Pairs of Equations

A **pair of equations** means two equations containing the same variables.
Usually, the variables are:

x and y

These two equations are solved together to find the values of the variables.

Example

$$x + y = 10$$

$$x - y = 2$$

These two equations together form a **pair of equations**.

General Form

$$ax + by = c$$

and

$$px + qy = r$$

There are 100 beads, black and white, in a box; 10 more black than white. How many black and how many white ?

Let the number of black beads be x .

Then the number of white beads is:

$$x - 10$$

Since the total number of beads is 100,

$$x + (x - 10) = 100$$

Simplifying,

$$2x - 10 = 100$$

Adding 10 on both sides,

$$2x = 110$$

Dividing by 2,

$$x = 55$$

Therefore,

- Number of black beads = 55
- Number of white beads = $55 - 10 = 45$

The price of a table and chair is 11000 rupees. The price of the table and four chairs is 14000 rupees. What's the price of each ?

Method 1: Using only One variable

Let the price of a chair be x rupees.

Then the price of the table is:

$$11000 - x$$

The total price of one table and four chairs is:

$$(11000 - x) + 4x$$

Simplifying,

$$11000 + 3x$$

This total is given as 14000 rupees.

Therefore,

$$11000 + 3x = 14000$$

Subtracting 11000 from both sides,

$$3x = 3000$$

Dividing by 3,

$$x = 1000$$

Hence,

•Price of one chair = 1000 rupees

Price of the table:

$$11000 - 1000 = 10000$$

Therefore,

•Price of the table = 10000 rupees.

Method 2: Using two Variable

Let the price of a chair be x rupees and the price of the table be y rupees.
Then, according to the problem:

$$\begin{aligned}x + y &= 11000 \\4x + y &= 14000\end{aligned}$$

From the first equation,

$$y = 11000 - x$$

Substituting this value of y in the second equation:

$$4x + (11000 - x) = 14000$$

Simplifying,

$$\begin{aligned}4x + 11000 - x &= 14000 \\11000 + 3x &= 14000\end{aligned}$$

Subtracting 11000 from both sides,

$$3x = 3000$$

Dividing by 3,

$$x = 1000$$

Therefore, the price of one chair is:

$$1000 \text{ rupees}$$

Substituting $x = 1000$ in

$$\begin{aligned}y &= 11000 - x \\y &= 11000 - 1000 \\y &= 10000\end{aligned}$$

Therefore, the price of the table is:

$$10000 \text{ rupees}$$

Of two numbers, the larger one is 5 times the smaller one; and subtracting the smaller from the larger gives 32. What are the numbers?

Method 1: Using only one variable

Let the smaller number be x .

Then the larger number is:

$$5x$$

According to the problem, subtracting the smaller number from the larger number gives 32.

Therefore,

$$5x - x = 32$$

Simplifying,

$$4x = 32$$

Dividing by 4,

$$x = 8$$

Therefore, the smaller number is:

$$8$$

The larger number is:

$$5 \times 8 = 40$$

Hence, the two numbers are:

$$8 \text{ and } 40$$

Method 2 : Using two variables

Let the smaller number be x and the larger number be y .

According to the problem:

The larger number is 5 times the smaller number.

$$y = 5x$$

Also, subtracting the smaller number from the larger number gives 32.

$$y - x = 32$$

Substituting $y = 5x$ in the second equation:

$$5x - x = 32$$

Simplifying,

$$4x = 32$$

Dividing by 4,

$$x = 8$$

Substituting $x = 8$ in

$$y = 5x$$

$$y = 5 \times 8$$

$$y = 40$$

Therefore,

•Smaller number = 8

•Larger number = 40

Hence, the two numbers are:

8 and 40

A fraction, after adding 1 to its numerator and simplifying, becomes $\frac{1}{2}$. If instead, 1 is added to the denominator and then simplified, it becomes $\frac{1}{3}$. Find the fraction.

Let the numerator be x and the denominator be y .

Therefore, the fraction is:

$$\frac{x}{y}$$

Condition 1

When 1 is added to the numerator, the fraction becomes:

$$\frac{1}{2}$$

So,

$$\frac{x+1}{y} = \frac{1}{2}$$

Cross multiplying,

$$\begin{aligned} 2(x+1) &= y \\ 2x+2 &= y \end{aligned}$$

Therefore,

$$y = 2x + 2$$

Condition 2

When 1 is added to the denominator, the fraction becomes: $\frac{1}{3}$

So,

$$\frac{x}{y+1} = \frac{1}{3}$$

Cross multiplying,

$$3x = y + 1$$

Therefore,

$$y + 1 = 3x$$

Substitute $y = 2x + 2$

$$(2x + 2) + 1 = 3x$$

$$2x + 3 = 3x$$

$$x = 3$$

Substituting $x = 3$ in

$$y = 2x + 2$$

$$y = 2(3) + 2$$

$$y = 6 + 2$$

$$y = 8$$

Answer

Therefore, the required fraction is:

$$\boxed{\frac{3}{8}}$$

Q1: Priya bought a bag and a pair of slippers for 1100 rupees. The bag costs 300 rupees more than the slippers. Find: the price of the slippers and the price of the bag

Solution Using Two Variables

Let the price of the slippers be x rupees.

Let the price of the bag be y rupees.

According to the question,

$$x + y = 1100$$

The bag costs 300 rupees more than the slippers.

So,

$$y = x + 300$$

Substituting $y = x + 300$ in the first equation,

$$x + (x + 300) = 1100$$

Simplifying,

$$2x + 300 = 1100$$

Subtracting 300 from both sides,

$$2x = 800$$

Dividing by 2,

$$x = 400$$

Therefore, the price of the slippers is:

$$400 \text{ rupees}$$

Now substitute $x = 400$ in

$$y = x + 300$$

$$y = 400 + 300$$

$$y = 700$$

Therefore, the price of the bag is:

$$700 \text{ rupees}$$

Answer

- Price of slippers = 400 rupees
- Price of bag = 700 rupees

Q2: The sum of two numbers is 26 and their difference is 4. Find the numbers.

Let the smaller number be x .

Let the larger number be y .

According to the question,

$$x + y = 26$$

$$y - x = 4$$

From the first equation,

$$y = 26 - x$$

Substituting this value of y in the second equation,

$$(26 - x) - x = 4$$

Simplifying,

$$26 - 2x = 4$$

Subtracting 26 from both sides,

$$-2x = -22$$

Dividing by -2 ,

$$x = 11$$

Substituting $x = 11$ in

$$y = 26 - x$$

$$y = 26 - 11$$

$$y = 15$$

Q3: The perimeter of a rectangle is 40 centimeters, and one side is 8 centimetres longer than the other. Calculate the lengths of the sides.

Let the shorter side be x centimetres.

Let the longer side be y centimetres.

The perimeter of a rectangle is:

$$2(l + b)$$

According to the question,

$$2(x + y) = 40$$

Dividing both sides by 2,

$$x + y = 20$$

Also, one side is 8 centimetres longer than the other.

$$y = x + 8$$

Substituting $y = x + 8$ in

$$x + y = 20$$

we get,

$$x + (x + 8) = 20$$

Simplifying,

$$2x + 8 = 20$$

Subtracting 8 from both sides,

$$2x = 12$$

Dividing by 2,

$$x = 6$$

Substituting $x = 6$ in

$$y = x + 8$$

$$y = 6 + 8$$

$$y = 14$$

Two equations

The price of 2 pens and 3 notebooks is 110 rupees. The price of 2 pens and 5 notebooks is 170 rupees. Find the price of a pen and a notebook.

Let the price of a pen be x rupees.

Let the price of a notebook be y rupees.

According to the question,

$$2x + 3y = 110$$

$$2x + 5y = 170$$

Subtract the first equation from the second equation.

$$(2x + 5y) - (2x + 3y) = 170 - 110$$

$$2x + 5y - 2x - 3y = 60$$

$$2y = 60$$

Dividing by 2,

$$y = 30$$

Therefore, the price of one notebook is:

$$30 \text{ rupees}$$

Substituting $y = 30$ in

$$2x + 3y = 110$$

$$2x + 3(30) = 110$$

$$2x + 90 = 110$$

$$2x = 20$$

$$x = 10$$

Therefore, the price of one pen is:

$$10 \text{ rupees}$$

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The price of 3 pencils and 4 pens is 66 rupees. The price of 6 pencils and 3 pens is 72 rupees. Find the price of each.

Let the price of one pencil be x rupees. Let the price of one pen be y rupees.

According to the question,

$$3x + 4y = 66$$

$$6x + 3y = 72$$

Eliminate x

Multiply the first equation by 2:

$$6x + 8y = 132$$

Second equation:

$$6x + 3y = 72$$

Subtract the second equation from the first equation:

$$(6x + 8y) - (6x + 3y) = 132 - 72$$

$$5y = 60$$

$$y = 12$$

Therefore, the price of one pen is:

12 rupees

Find x

Substitute $y = 12$ in

$$3x + 4y = 66$$

$$3x + 4(12) = 66$$

$$3x + 48 = 66$$

$$3x = 18$$

$$x = 6$$

Therefore, the price of one pencil is:

6 rupees

Answer

- Price of one pencil = 6 rupees
- Price of one pen = 12 rupees

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The price of 4 pens and 3 pencils is 66 rupees. The price of 7 pens and 3 pencils is 111 rupees. Find the price of a pen and a pencil.

Let the price of one pen be x rupees.

Let the price of one pencil be y rupees.

According to the question,

$$4x + 3y = 66$$

$$7x + 3y = 111$$

Subtract the first equation from the second equation:

$$(7x + 3y) - (4x + 3y) = 111 - 66$$

$$7x + 3y - 4x - 3y = 45$$

$$3x = 45$$

$$x = 15$$

Therefore, the price of one pen is:

15 rupees

Find the price of a pencil

Substitute $x = 15$ in

$$4x + 3y = 66$$

$$4(15) + 3y = 66$$

$$60 + 3y = 66$$

$$3y = 6$$

$$y = 2$$

Therefore, the price of one pencil is:

2 rupees

Answer

- Price of one pen = 15 rupees
- Price of one pencil = 2 rupees

Master X's Academy

The perimeter of a rectangle is 26 centimetres. Another rectangle with twice the length and thrice the breadth has perimeter 62 centimetres. Find the length and breadth of the first rectangle.

Let the length of the first rectangle be x cm.

Let the breadth of the first rectangle be y cm.

The perimeter of a rectangle is:

$$2(l + b)$$

According to the question,

$$2(x + y) = 26$$

Dividing by 2,

$$x + y = 13 \dots (1)$$

For the second rectangle:

•Length = $2x$

•Breadth = $3y$

Its perimeter is 62 cm.

$$2(2x + 3y) = 62$$

Dividing by 2,

$$2x + 3y = 31 \dots (2)$$

From equation (1),

$$x + y = 13$$

$$x = 13 - y$$

Substituting this value of x in equation (2),

$$2(13 - y) + 3y = 31$$

$$26 - 2y + 3y = 31$$

$$26 + y = 31$$

$$y = 5$$

Substituting $y = 5$ in equation (1),

$$x + y = 13$$

$$x + 5 = 13$$

$$x = 8$$

Answer

Therefore,

- Length of the first rectangle = 8cm
- Breadth of the first rectangle = 5cm

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Master X's Academy

When a small vessel was filled and emptied five times and a big vessel two times into a bucket, the bucket contained 20 litres.
Instead, when the small vessel was filled and emptied twice and the big vessel three times, the bucket contained 19 litres. Find the capacity of each vessel.

Let the capacity of the small vessel be x litres.

Let the capacity of the big vessel be y litres.

According to the question,

$$5x + 2y = 20 \dots (1)$$

$$2x + 3y = 19 \dots (2)$$

Eliminate y

Multiply equation (1) by 3:

$$15x + 6y = 60 \dots (3)$$

Multiply equation (2) by 2:

$$4x + 6y = 38 \dots (4)$$

Subtract equation (4) from equation (3):

$$(15x + 6y) - (4x + 6y) = 60 - 38$$

$$15x + 6y - 4x - 6y = 22$$

$$11x = 22$$

$$x = 2$$

Therefore, the capacity of the small vessel is:

2 litres

Find y

Substitute $x = 2$ in equation (1):

$$5x + 2y = 20$$

$$5(2) + 2y = 20$$

$$10 + 2y = 20$$

$$2y = 10$$

$$y = 5$$

Therefore, the capacity of the big vessel is:

5 litres

Answer

- Capacity of the small vessel = 2litres
- Capacity of the big vessel = 5litres

MASTER X'S ACADEMY

The price of two kilograms of orange and three kilograms of apple is 520 rupees.
The price of three kilograms of orange and two kilograms of apple is 480 rupees. Find the price of each.

Let the price of one kilogram of orange be x rupees.

Let the price of one kilogram of apple be y rupees.

According to the question,

$$2x + 3y = 520 \dots (1)$$

$$3x + 2y = 480 \dots (2)$$

Eliminate x

Multiply equation (1) by 3:

$$6x + 9y = 1560 \dots (3)$$

Multiply equation (2) by 2:

$$6x + 4y = 960 \dots (4)$$

Subtract equation (4) from equation (3):

$$(6x + 9y) - (6x + 4y) = 1560 - 960$$

$$6x + 9y - 6x - 4y = 600$$

$$5y = 600$$

$$y = 120$$

Therefore, the price of one kilogram of apple is:

120 rupees

Find x

Substitute $y = 120$ in equation (1):

$$\begin{aligned}2x + 3y &= 520 \\2x + 3(120) &= 520 \\2x + 360 &= 520 \\2x &= 160 \\x &= 80\end{aligned}$$

Therefore, the price of one kilogram of orange is:

80 rupees

Answer

- Price of one kilogram of orange = 80 rupees
- Price of one kilogram of apple = 120 rupees

MASTER X'S ACADEMY

Master X's Academy

Four years ago, Rahim's age was three times the age of Ramu. After two years, Rahim's age will become two times the age of Ramu. Find their present ages.

Let the present age of Rahim be x years.

Let the present age of Ramu be y years.

Four years ago

Rahim's age:

$$x - 4$$

Ramu's age:

$$y - 4$$

According to the question,

$$x - 4 = 3(y - 4)$$

$$x - 4 = 3y - 12$$

$$x - 3y = -8 \dots (1)$$

After two years

Rahim's age:

$$x + 2$$

Ramu's age:

$$y + 2$$

According to the question,

$$x + 2 = 2(y + 2)$$

$$x + 2 = 2y + 4$$

$$x - 2y = 2 \dots (2)$$

Eliminate x

Subtract equation (2) from equation (1):

$$\begin{aligned}(x - 3y) - (x - 2y) &= -8 - (-2) \\ x - 3y - x + 2y &= -8 - 2 \\ -y &= -10 \\ y &= 10\end{aligned}$$

Therefore, the present age of Ramu is:

10 years

Find x

Substitute $y = 10$ in equation (2):

$$\begin{aligned}x - 2(10) &= -2 \\ x - 20 &= -2 \\ x &= 18\end{aligned}$$

Therefore, the present age of Rahim is:

18 years

Answer

- Present age of Rahim = 18 years
- Present age of Ramu = 10 years

Four times a number added to three times another number gives 43. Two times the second number subtracted from three times the first gives 11. Find the numbers.

Let the first number be x and the second number be y .

Writing the given information as equations, we get

$$4x + 3y = 43 \dots (1)$$

$$3x - 2y = 11 \dots (2)$$

As before, we multiply Equation (1) by 3 and Equation (2) by 4 to make the coefficients of x the same.

$$(1) \times 3: 12x + 9y = 129 \dots (3)$$

$$(2) \times 4: 12x - 8y = 44 \dots (4)$$

Now subtracting equation (4) from equation (3),

$$(12x + 9y) - (12x - 8y) = 129 - 44$$

$$12x + 9y - 12x + 8y = 85$$

$$17y = 85$$

$$y = 5$$

Using $y = 5$ in equation (1),

$$4x + 3(5) = 43$$

$$4x + 15 = 43$$

$$4x = 28$$

$$x = 7$$

Thus, the first number is:

7

and the second number is:

5

The sum of two numbers is 28 and their difference is 12. What are the numbers ?

Let the first number be x and the second number be y .

Writing the given information as equations, we get

$$x + y = 28 \dots (1)$$

$$x - y = 12 \dots (2)$$

Adding equation (1) and equation (2),

$$(x + y) + (x - y) = 28 + 12$$

$$x + y + x - y = 40$$

$$2x = 40$$

$$x = 20$$

Using $x = 20$ in equation (1),

$$20 + y = 28$$

$$y = 8$$

Thus, the first number is:

$$20$$

and the second number is:

$$8$$

Master X's Academy

General Principle

Let the larger number be x and the smaller number be y .

Then,

•Sum of the numbers:

$$x + y$$

•Difference of the numbers:

$$x - y$$

Adding the Sum and Difference

$$(x + y) + (x - y)$$

Removing brackets,

$$x + y + x - y$$
$$2x$$

Thus,

$$(x + y) + (x - y) = 2x$$

So, **the sum and difference of two numbers added together gives twice the larger number.**

Subtracting the Difference from the Sum

$$(x + y) - (x - y)$$

Removing brackets,

$$x + y - x + y$$
$$2y$$

Thus,

$$(x + y) - (x - y) = 2y$$

So, **subtracting the difference from the sum gives twice the smaller number.**

The difference of the two smaller angles of a right triangle is 20° . Calculate all three angles.

Let the smaller angle be x° .

Let the other angle be y° .

In a right triangle, one angle is always:

$$90^\circ$$

Therefore, the sum of the other two angles is also:

$$90^\circ$$

So,

$$x + y = 90 \dots (1)$$

The difference of the two smaller angles is 20° .

$$y - x = 20 \dots (2)$$

Adding equation (1) and equation (2),

$$(x + y) + (y - x) = 90 + 20$$

$$x + y + y - x = 110$$

$$2y = 110$$

$$y = 55$$

Using $y = 55$ in equation (1),

$$x + 55 = 90$$

$$x = 35$$

Answer

Therefore, the three angles of the triangle are:

$$35^\circ, 55^\circ, 90^\circ$$

The price of 17 trophies and 16 medals is 2180 rupees. The price of 16 trophies and 17 medals is 2110 rupees. What is the price of each ?

Let the price of one trophy be x rupees. Let the price of one medal be y rupees.

According to the question,

$$17x + 16y = 2180 \dots (1)$$

$$16x + 17y = 2110 \dots (2)$$

Eliminate y

Multiply equation (1) by 17:

$$289x + 272y = 37060 \dots (3)$$

Multiply equation (2) by 16:

$$256x + 272y = 33760 \dots (4)$$

Subtract equation (4) from equation (3):

$$(289x + 272y) - (256x + 272y) = 37060 - 33760$$

$$289x + 272y - 256x - 272y = 3300$$

$$33x = 3300$$

$$x = 100$$

Therefore, the price of one trophy is:

100 rupees

Find y

Substitute $x = 100$ in equation (1):

$$17(100) + 16y = 2180$$

$$1700 + 16y = 2180$$

$$16y = 480$$

$$y = 30$$

Therefore, the price of one medal is:

30 rupees

Answer

- Price of one trophy = 100 rupees
- Price of one medal = 30 rupees

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